

# Project Report

## 1. Project Title

### Online Bookstore

#### Abstract

The Online Bookstore is a web-based application developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js) that allows users to conveniently browse, search, and purchase books online. The system provides an intuitive platform for managing book inventories and enhancing the overall book-buying experience.

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## 2. Problem Statement and Objectives

### Problem Statement

Buying books through physical stores can be time-consuming and inconvenient, especially when users need to search across multiple shops for availability, pricing, and categories. Manual book purchasing lacks quick comparison, personalized recommendations, and efficient order tracking.

### Objectives

- To design and develop a user-friendly online bookstore system
  - To allow users to browse, search, and purchase books easily
  - To provide categorized listings and detailed book information
  - To enable secure user authentication and order management
  - To ensure reliable data storage for users, books, and transactions
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## 3. System Architecture and Workflow

### System Architecture

The application follows a three-tier architecture:

- **Frontend:** React.js , Tailwind CSS
- **Backend:** Node.js with Express.js
- **Database:** MongoDB

The frontend communicates with the backend through RESTful APIs. The backend processes requests, handles authentication, and interacts with the database.

### Workflow

1. User registers or logs into the system
2. User browses or searches for books by category or title

3. User views book details and adds selected books to the cart
  4. Order and user data are stored in MongoDB
  5. Backend processes orders and manages inventory
  6. Frontend displays order status and purchase history.
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## 4. Technologies Used

- **Frontend:** React.js, Tailwind CSS
  - **Backend:** Node.js, Express.js
  - **Database:** MongoDB
  - **Authentication:** JSON Web Token (JWT)
  - **Tools:** Git, GitHub, VS Code, Postman
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## 5. Module-Wise Description

### 5.1 Authentication Module

- User registration and login
- JWT-based authentication

### 5.2 Book Management Module

- Add, update, delete books
- Categorize and Search books

### 5.3 Shopping Cart & Order Module

- Manage Cart Items and Order Processing

### 5.4 Payment & Shipping Module

- Payment Gateway Integration
- Order Tracking

### 5.5 Analytics & Reporting Module

- Sales Analysis and Inventory Reports
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## 6. Database Design

The database is designed using MongoDB, a NoSQL document-oriented database, which provides flexibility, scalability, and high performance for handling user and transaction data.

## 6.1 Database Overview

The application uses MongoDB collections to store structured data in JSON-like documents. Each user has associated purchase orders and a shopping cart, enabling efficient querying, inventory management, and sales analytics.

## 6.2 Collections and Schema Design

**User Collection** This collection stores user authentication and profile details.

- **\_id**: Unique identifier generated by MongoDB
- **name**: Full name of the user
- **email**: Unique email address used for login
- **password**: Encrypted password (hashed)
- **role**: User role (e.g., Customer or Admin)
- **createdAt**: Account creation timestamp

### Purpose:

- Manages user identity and authentication
- Ensures secure access to personal order history and cart data

**Book Collection** This collection stores details of all books available in the inventory.

- **\_id**: Unique book identifier
- **title**: Title of the book
- **author**: Name of the author
- **category**: Genre of the book (e.g., Fiction, Sci-Fi)
- **price**: Cost per unit
- **stock**: Quantity available in inventory
- **image**: URL link to book cover image

### Purpose:

- Manages the product catalog for browsing and searching
- Tracks inventory levels to prevent overselling

**Order Collection** This collection stores all purchase records and transaction details.

- **\_id**: Unique order identifier
- **userId**: Reference to the User collection
- **books**: Array of objects containing Book IDs and quantities purchased
- **totalAmount**: Final monetary value of the order
- **status**: Current order status (e.g., Processing, Shipped)

- **date:** Date of purchase

**Purpose:**

- Stores transaction records linked to users
- Stores transaction records linked to users

### 6.3 Relationships

- **One-to-Many relationship between User and Order:** A single user can have multiple orders.
- **One-to-Many relationship between Book and Order:** A single order contains multiple books

### 6.4 Data Security 🔒

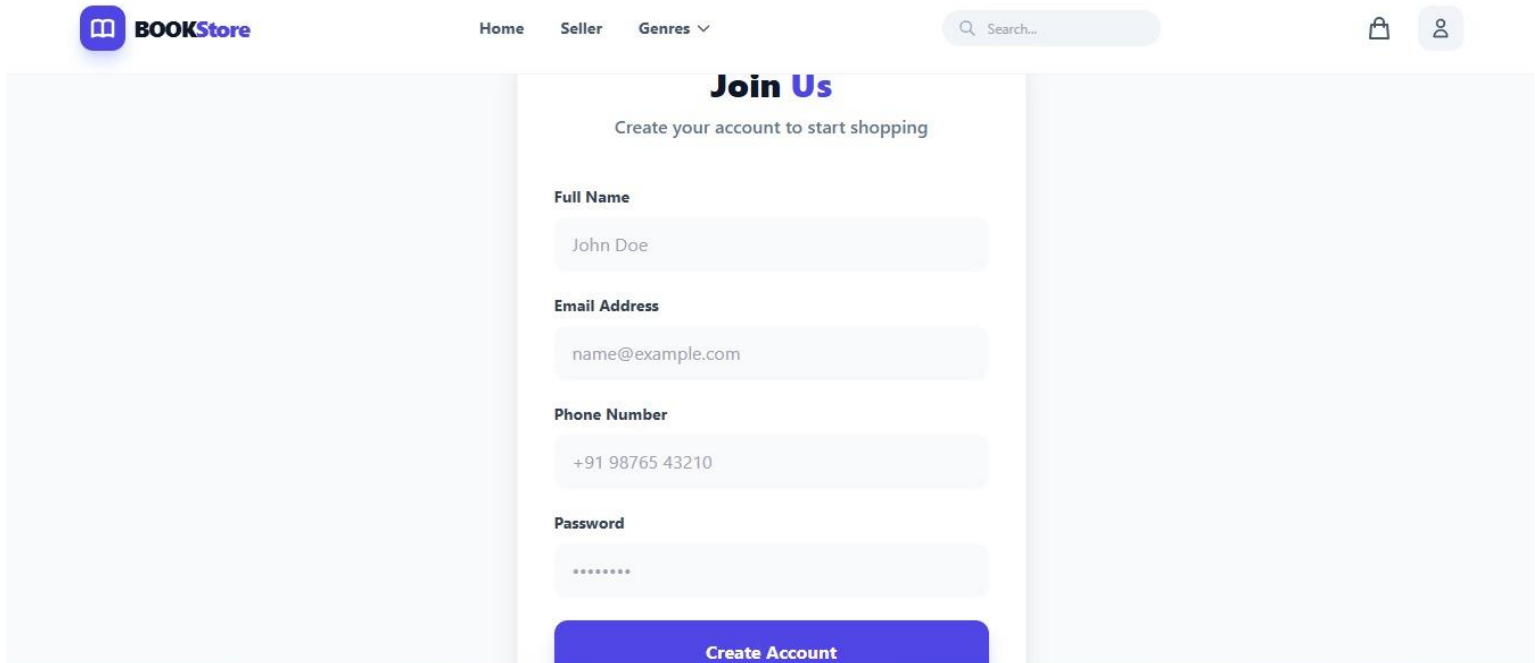
- Passwords are stored using hashing techniques (e.g., bcrypt) to prevent unauthorized access.
  - User-specific data access ensured through JWT authentication, protecting personal order history and payment details.
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## 7. Individual Roles and Responsibilities (10 Members)

1. [Vivek BARNWAL](#) – Project Lead & Github Directory
  2. [SUBHABRATA PAUL](#)– Frontend Development(order section + Deployment)
  3. [SAKET RAI](#) – Backend Development(Auth + Admin+Routes)
  4. [SAMEER KUMAR AMBASTH](#)– Database Design &Documentation
  5. [VIBHASH KUMAR](#)– Frontend Development(Admin)
  6. [SHASHANK MISHRA](#)– Frontend Development(pages)
  7. [SWAPNANIL DAS](#)– Backend Development(controller)
  8. [SAYAK BAG](#)– Backend Development(Server+Middleware)
  9. [SAYANTAN MAJI](#)– Testing & Debugging
  10. [SAKIR MALLIK](#) – Testing & Debugging
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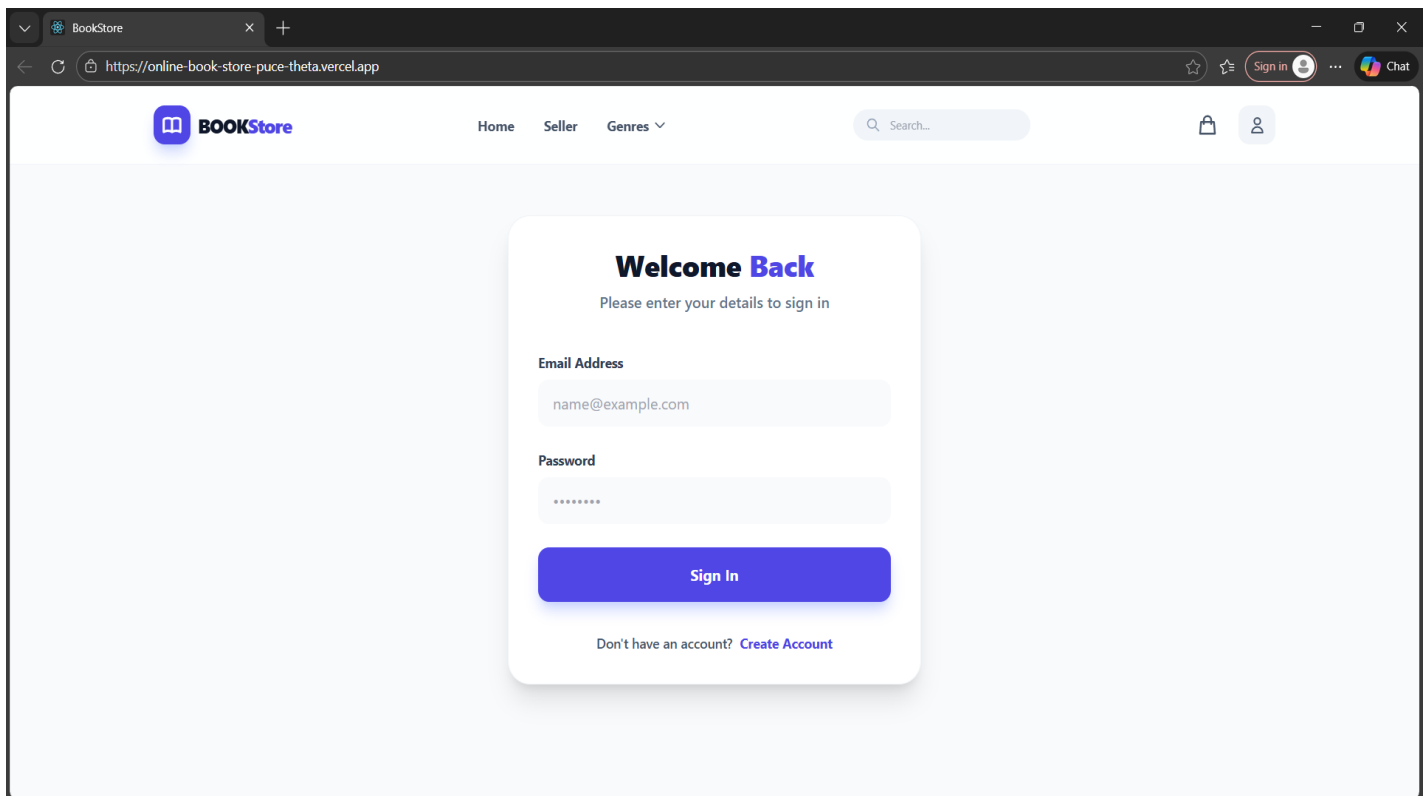
## 8. Results, Screenshots, and Output

- User authentication:-



The screenshot shows the 'Join Us' registration page of the BOOKStore application. The page has a light blue header with the BOOKStore logo, navigation links (Home, Seller, Genres), a search bar, and icons for a shopping bag and user profile. The main content area features a white card with the title 'Join Us' and the subtitle 'Create your account to start shopping'. The form includes four input fields: 'Full Name' (with the placeholder 'John Doe'), 'Email Address' (with the placeholder 'name@example.com'), 'Phone Number' (with the placeholder '+91 98765 43210'), and 'Password' (with a masked placeholder '\*\*\*\*\*'). A blue 'Create Account' button is positioned at the bottom of the form.

- User Sign in:-



The screenshot shows the 'Welcome Back' sign-in page of the BOOKStore application. The page has a light blue header with the BOOKStore logo, navigation links (Home, Seller, Genres), a search bar, and icons for a shopping bag and user profile. The main content area features a white card with the title 'Welcome Back' and the subtitle 'Please enter your details to sign in'. The form includes two input fields: 'Email Address' (with the placeholder 'name@example.com') and 'Password' (with a masked placeholder '\*\*\*\*\*'). A blue 'Sign In' button is positioned at the bottom of the form. Below the button, there is a link that says 'Don't have an account? [Create Account](#)'.

- Admin login:-

The screenshot shows the 'Seller Portal' login page. At the top, there's a navigation bar with 'Home', 'Seller', and 'Genres' links, a search bar, and a 'Sign in' button. The main content area features a 'Seller Portal' header with a shopping bag icon and the text 'Authorized Access Only'. Below this, there are input fields for 'Admin Email' and 'Password'. A 'Login to Dashboard' button is prominently displayed. A link to 'Back to Marketplace' is located below the login button. At the bottom, a 'DEMO ACCESS' section provides a pre-filled email 'admin@gmail.com' and password 'admin', with an 'AUTO FILL' button next to it.

BOOKStore

Home Seller Genres

Search...

**Seller Portal**  
Authorized Access Only

Admin Email

Enter admin email

Password

\*\*\*\*\*

**Login to Dashboard**

[Back to Marketplace](#)

DEMO ACCESS

Email: admin@gmail.com

Password: admin

**AUTO FILL**

Seller Portal:-

The screenshot displays the 'Add New Book' form. The form has a blue header with the title 'Add New Book' and the instruction 'Fill in the details to list a new book'. The form fields are organized into two columns: 'TITLE' and 'AUTHOR' in the first row, 'PRICE (RS.)' and 'GENRE' in the second row, 'STOCK' and 'IMAGE URL' in the third row, and a single 'DESCRIPTION' field at the bottom. Each field is represented by a light gray input box. A vertical scrollbar is visible on the right side of the form.

**Add New Book**  
Fill in the details to list a new book

TITLE

AUTHOR

PRICE (RS.)

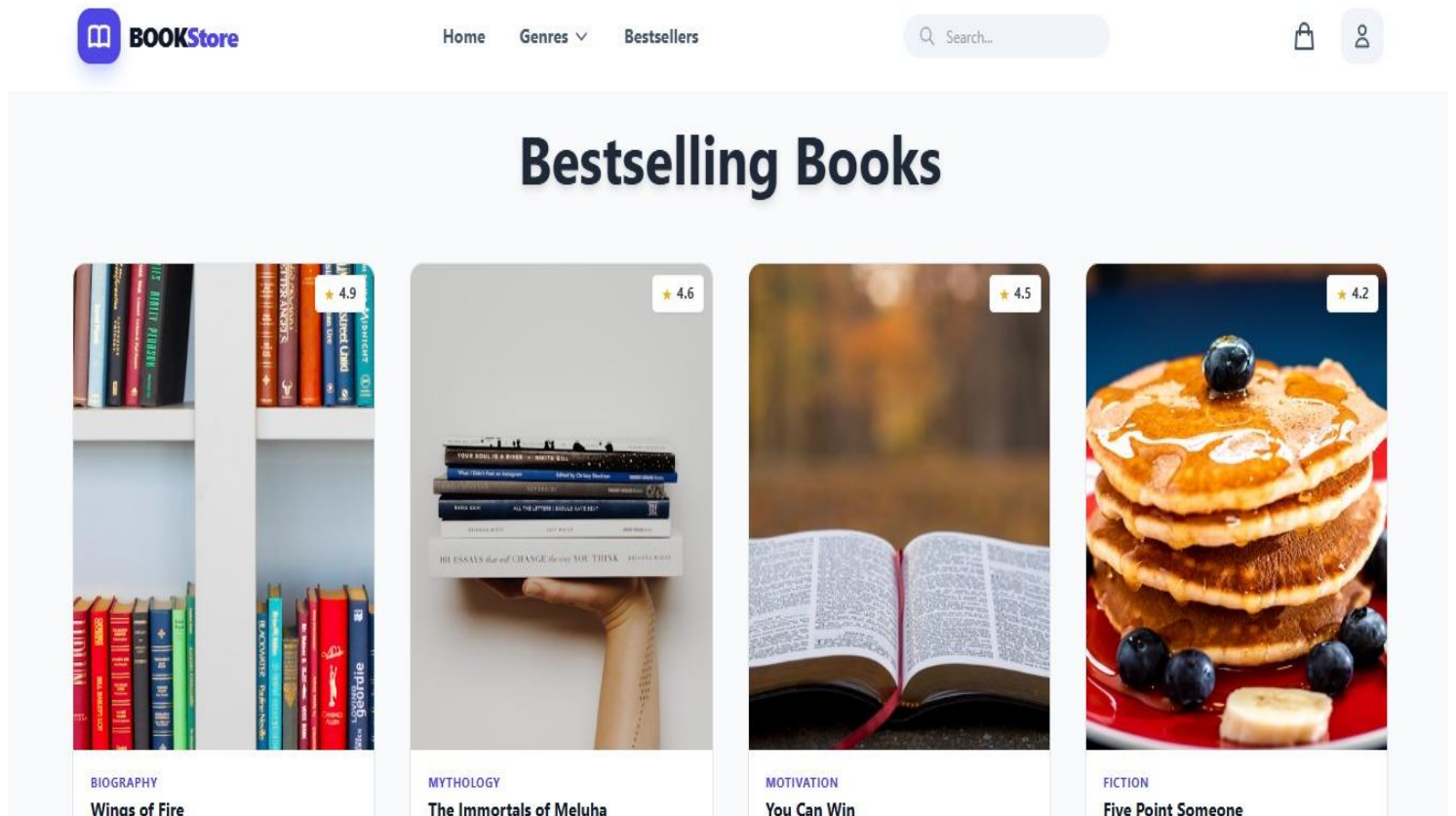
GENRE

STOCK

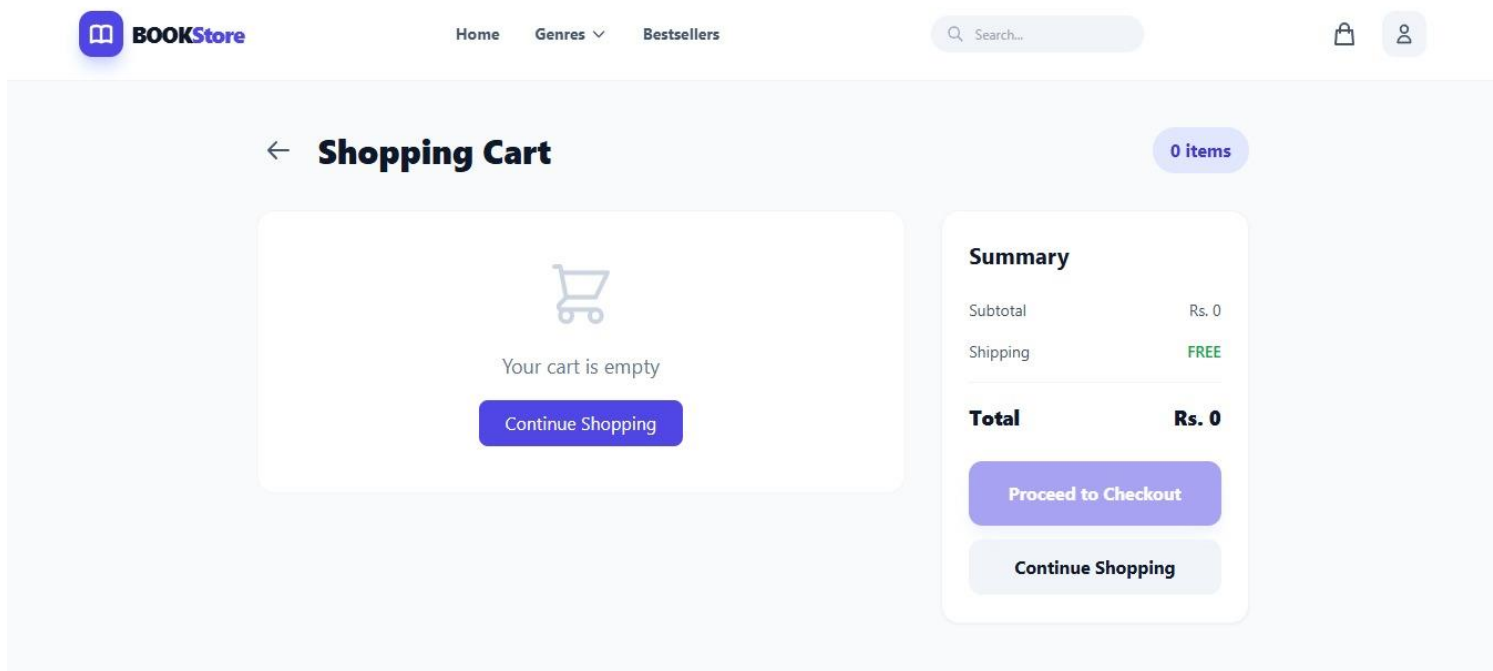
IMAGE URL

DESCRIPTION

## .Website Homepage:-



## • Shopping Cart:-



- **Seller Order Management:-**

[Dashboard](#)

[Inventory](#)

[Orders](#)

## Orders

Welcome back, Demo Admin! Manage your store inventory.

### Order Management

Monitor and manage all customer orders

| ORDER ID | CUSTOMER                                              | DATE      | AMOUNT          | STATUS  | ACTIONS                      |
|----------|-------------------------------------------------------|-----------|-----------------|---------|------------------------------|
| #187412  | <b>Subhabrata Paul</b><br>subhabratapaul277@gmail.com | 1/28/2026 | <b>Rs. 1920</b> | PENDING | <a href="#">View Details</a> |

- **Deployment:-**

DeploymentLogsResourcesSourceOpen Graph

Created

Subha12125 10m ago

Status

Ready Latest

Duration

34s 10m ago

Environment

Production Current

Domains

online-book-store-puce-theta.vercel.app +1

online-book-store-git-main-subhabratapaul277-gmailcoms-projects.vercel.app

online-book-store-fmcm5xtg.vercel.app

Source

main

ec831f9 Merge pull request #17 from Subha12125/subha-01

> Deployment Settings

4 Recommendations

Build Logs

34s

89 lines

Find in logs

Ctrl F

20:36:32.040 npm install -g serve

20:36:32.040 serve -s build

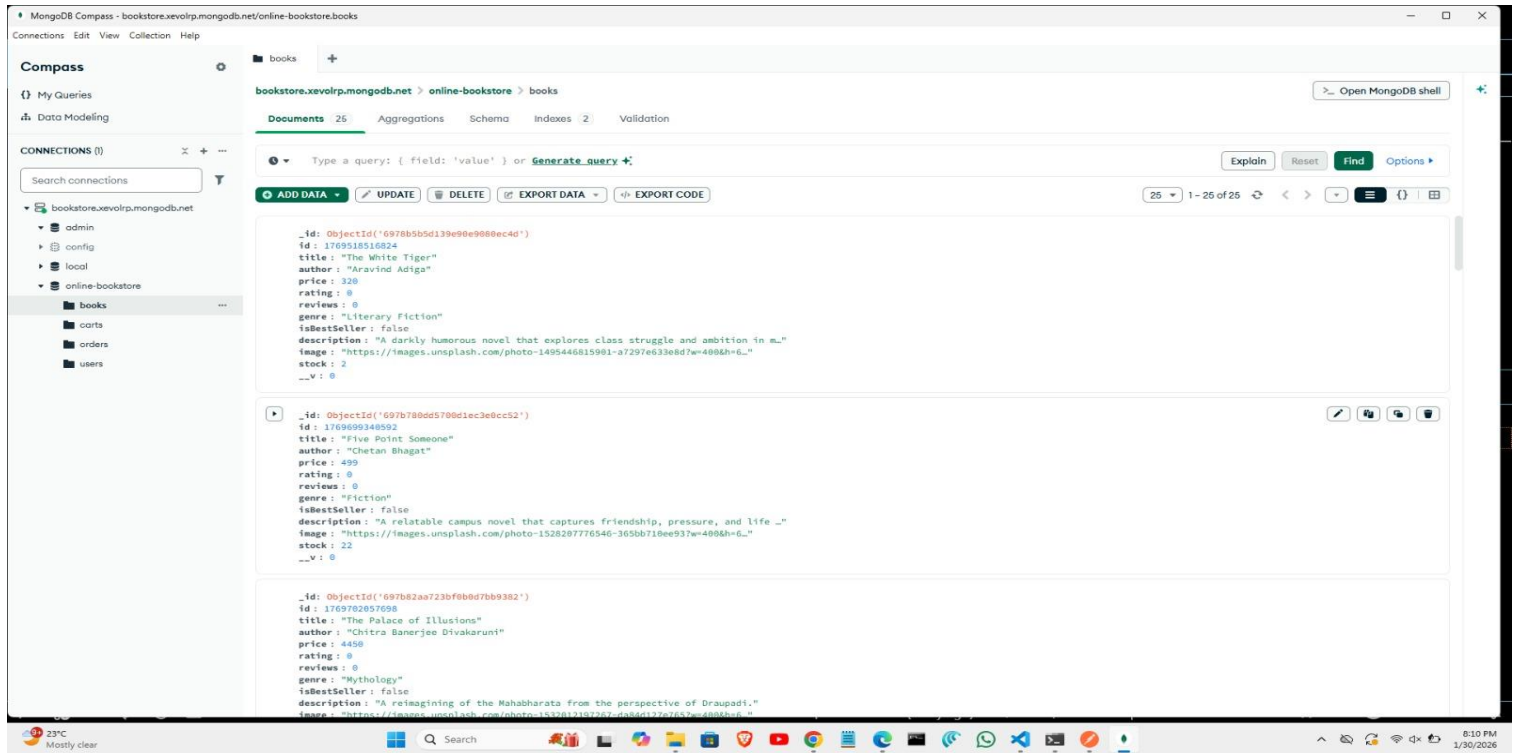
20:36:32.040

20:36:32.040 Find out more about deployment here:

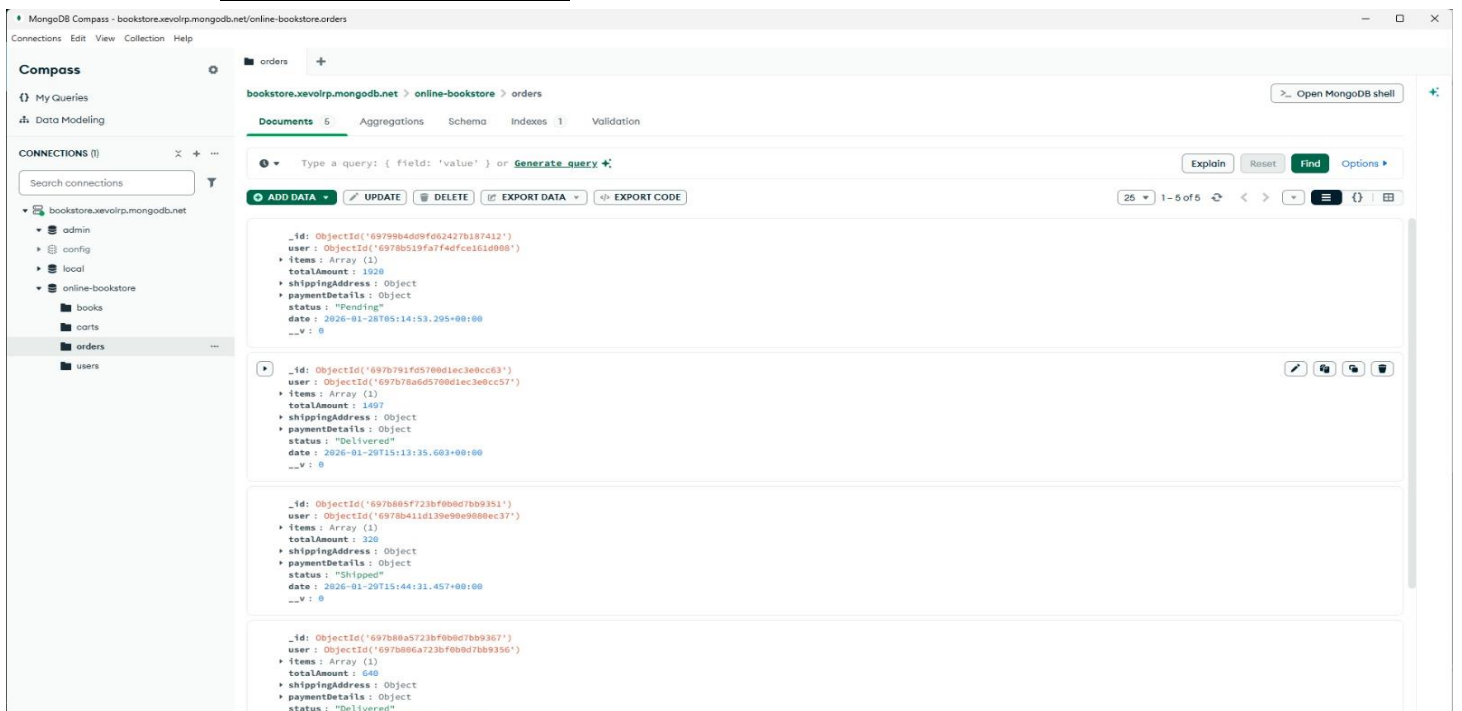


## OUTPUT(Stored in Mongodb atlas):-

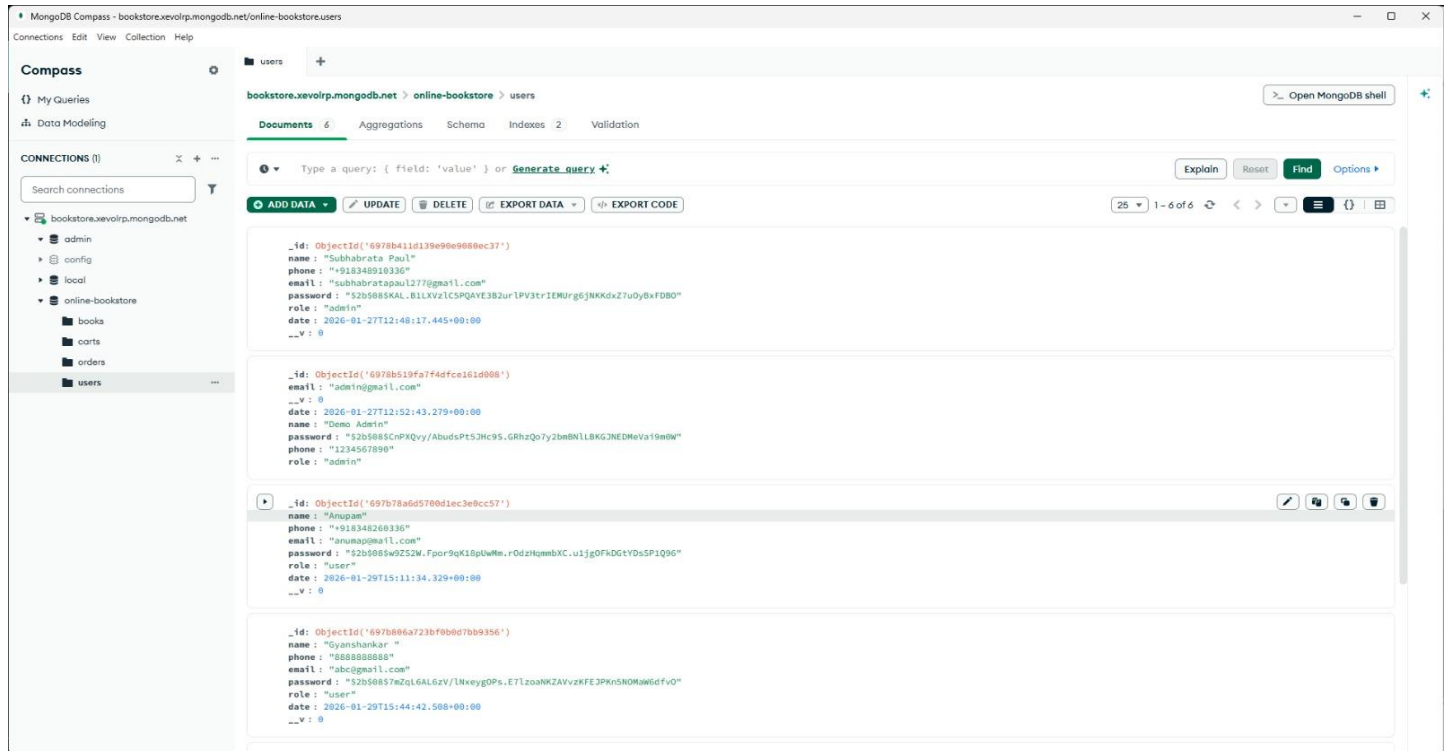
### • Database of Books :-



### • Database of Order:-



## • Database of users:-



## 9. Conclusion and Future Scope

### Conclusion

The "Online Bookstore" project has been successfully developed as a robust full-stack web application, effectively bridging the gap between users and digital book purchasing.

### Future Scope

While the current application is fully functional, the following enhancements can be implemented to scale the project for a commercial environment:

#### 1. Payment Gateway Integration

- The current checkout process simulates payment by accepting card numbers without processing them through a banking network

#### 2. Advanced Search & Filtering

- The current getBooks function retrieves the full list of books.
- Future updates can implement server-side pagination and advanced filtering to improve performance with large datasets.

#### 3. Enhanced Security Features

- Although the User model includes fields for password reset tokens the application can be further secured by implementing **Two-Factor Authentication (2FA)** via the stored phone number field.

#### 4. **Automated Order Tracking**

- Currently, orders are created and stored.

#### 5. **Review and Rating System**

- To enhance user engagement, a feedback mechanism allowing users to rate and review specific books can be linked to the book details schema.